BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ABBOTTABAD

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023 • Paper Code: 10231

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12

Note: Attempt all questions of Section-A by filling the corresponding bubble on the MCQs RESPONSE SHEET. It is mandatory to return the attempted MCQs sheet to the Superintendent within given time.

Q1. The coldest region in the atmosphere is .....

(A) troposphere
(B) stratosphere
(C) mesosphere
(D) thermosphere

Q2. Plants convert glucose into .....

(A) amino acids
(B) lipids
(C) proteins
(D) vitamins

Q3. The impurity present in mineral is called .....

(A) metal
(B) ore
(C) gangue
(D) salt

Q4. Matte is the mixture of .....

(A) Cus and FeS
(B) Cu_2S and FeS
(C) Cu_2O and FeS
(D) Cu_2O and FeO

Q5. Which one of the following is the formula of baking soda?

(A) $NaSO_4$
(B) Na_2CO_3
(C) NaCl
(D) $NaHCO_3$

Q6. Pick up the inorganic substance from the following.

(A) CH_4
(B) CH_3OH
(C) NaCN
(D) CH_3Cl

Q7. Which one is least reactive?

(A) ethyne
(B) propene
(C) ethene
(D) Ethane

Q8. Pick up the disaccharide from the following.

(A) glucose
(B) fructose
(C) sucrose
(D) starch

Q9. The density of water is maximum at .....

(A) −40°C
(B) 40°C
(C) −4°C
(D) 4°C

Q10. Active mass means .....

(A) total mass of reactants
(B) total mass of products
(C) total mass of products and reactants
(D) concentration of reactants and products in mole per dm^3

Q11. Neutral solution has a pH value of .....

(A) 3
(B) 5
(C) 7
(D) 14

Q12. Oxygen is attached on both sides to carbon atom in .....

(A) alcohol
(B) ketone
(C) aldehyde
(D) ether

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 32

Q1. Attempt any eight of the following short questions. Each question carries 4 marks

(i) Mention any two ways to recognize equilibrium.

(ii) How can salt be prepared by the action of acid and base?

(iii) Write down four uses of organic compounds.

(iv) What happens when alkyl halide is reduced?

(v) Differentiate between simple and complex lipids.

(vi) Briefly discuss how acid rain affects the aquatic life.

(vii) Water is a universal solvent. Why?

(viii) Compare inorganic and organic origin of petroleum.

(ix) Write down four characteristics of homologous series.

(x) Vitamins are vital for us. Why?

(xi) Mention the names and chemical formulas of four salts used in daily life.

SECTION-C

Maximum Marks: 21

Note: Attempt any three of the following questions. Each question carries 7 marks

Q2.

(i) Define the auto-ionization of water and find out the pH of water.

(ii) Calculate the pH and pOH of 0.001M solution of nitric acid (HNO_3).

Q3.

(i) Explain functional groups. Give function group of two different organic compounds.

(ii) Write down three physical properties of alkenes.

Q4.

(i) Describe the nature of bonding in proteins with proper diagram.

(ii) Write down three effects of air pollution.

Q5.

(i) Differentiate between temporary hard water and permanent hard water.

(ii) Mention three raw materials for the manufacture of sodium carbonate with proper names and chemical formulas.